

Q10 Ans: B

Total energy = maximum KE

$$KE_{\max} = \frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2}m\left(\frac{2\pi}{T}\right)^2x_0^2$$

$$4.0 \times 10^{-3} = \frac{1}{2}(0.20)\left(\frac{2\pi}{T}\right)^2(0.050)^2$$

$$T = 1.6 \text{ s}$$